FedIRT-DP with Inaccurate Response Screening

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Overview

The FedIRT-DP model developed by Zhou, Luo & Ji (2025) is a Federated Learning approach applied to Item Response Theory (IRT) models with an added student-level Differential Privacy (DP) layer to further obfuscate adversarial attempts to reverse-engineer student data. On top of effectively anonymizing student data, the DP layer is also shown to be capable of mitigating the impact of extreme response rows, as shown in Study 3.

The primary motivation behind this report was how FedIRT, despite its robustness against extreme response rows with DP, possesses no counter-measures against such rows. This motivated me to look into the potential benefits of an implementation that could identify such response rows from the individual student gradients.

A brief initial check proved that Local Outlier Factor (LOF) and Angle-Based Outlier Detection (ABOD) algorithms were able to identify the rows with extreme responses. However, before further examination, two preceding questions had to be addressed: Is the impact of extreme response rows significant enough to risk false positives from an additional statistical model to detect them? And how does FedIRT fare against other forms of inaccurate responses?

Stress-Testing FedIRT Against Inaccurate Responses

Procedure

Staying faithful to the study, all parameters were followed if explicitly stated; by the very least inferred if the paper implied so. Specifically, school effect was set to $s_k = 0$, with each of the 10 schools ($K = 10$) having 100 students each ($N_k = 100$). The test given to each school was dichotomously scored with 10 items ($J = 10$),

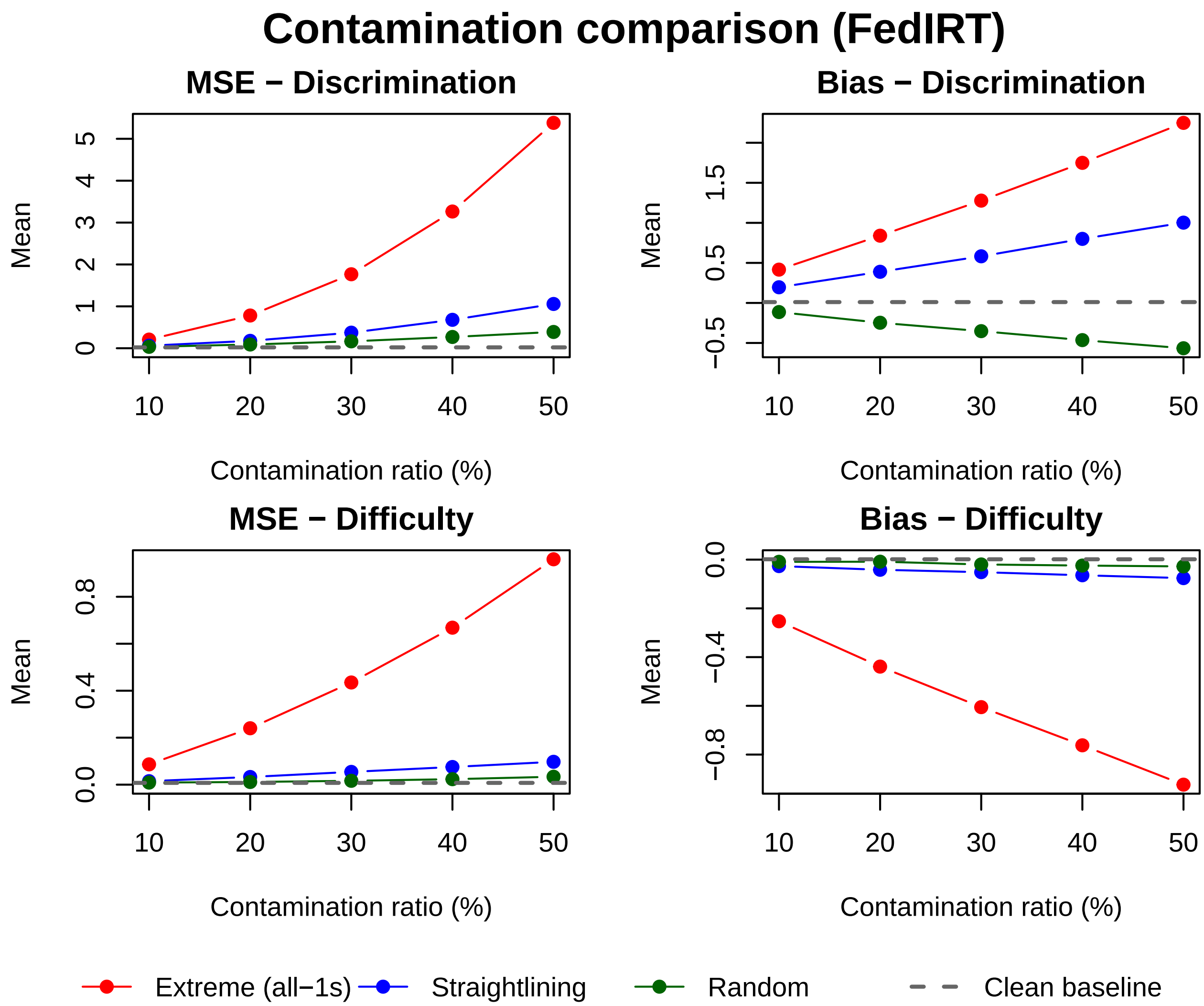
Individual abilities for each student were generated from a standard normal distribution ($\theta_i \sim N(0, 1)$) and fixed. Discrimination parameters were drawn from a uniform distribution ranging from $\alpha \in [0.5, 2]$, and difficulty parameters were drawn from a uniform distribution ranging from $\beta \in [-1, 1]$ to encompass both low and high parameter conditions. Student responses were generated using the 2PL Model probability using these fixed parameters.

The contamination conditions examined were straightlining (a certain number of items in a row are answered the same), and random reporting. Extreme response rows (all-ones) was also included to ensure the setup was consistent with the original study. As inferred from the plots in Study 3, a range of contaminated response proportions were tested from $[0.1, 0.2, ..., 0.5]$ for each contamination condition. Each proportion and each condition was repeated 100 times ($T = 100$).

The same MSE and bias formulas (Eq. 26) were used from the study for each proportion and contamination condition.

Results

The behavior of the MSE and bias of the difficulty and discrimination parameters are consistent with those from the original study for the extreme response condition. MSE and bias tend to increase as the proportion of responses consisting of all ones increase, except for the difficulty parameter which slightly decreases.



The MSE and bias of the other conditions, straightlining and random reporting, tend to follow the same positive relationship with the proportion of inaccurate responses but to a lesser degree. An exception to this is the bias of the discrimination parameter for random reporting, which steadily decreases below the baseline levels.

The severity of the damage done to the MSE and bias of the parameters visibly depends on the number of rows that are similar to extreme value responses. The more rows there are that consist of a single repeated response, the more MSE and bias are affected. Given that random responding is the least similar to this pattern, it has the most distinct bias and MSE values from the extreme response cases- with partial straightlining sitting in between them as the middle ground between the two as per its definition.

Table 1: MSE by contamination type and proportion

	Discrimination (a)				**Difficulty (b)**			
Prop (%)	Clean	Extreme	Straight	Random	Clean	Extreme	Straight	Random
10	0.021	0.207	0.061	0.034	0.007	0.086	0.015	0.009
20	0.021	0.781	0.177	0.089	0.007	0.240	0.033	0.011
30	0.021	1.766	0.372	0.165	0.007	0.435	0.054	0.016
40	0.021	3.264	0.678	0.269	0.007	0.668	0.075	0.023
50	0.021	5.380	1.057	0.389	0.007	0.960	0.097	0.034

Table 2: Bias by contamination type and proportion

	Discrimination (a)				Difficulty (b)			
Prop (%)	Clean	Extreme	Straight	Random	Clean	Extreme	Straight	Random
10	0.011	0.416	0.196	-0.114	0.001	-0.253	-0.027	-0.009
20	0.011	0.840	0.389	-0.247	0.001	-0.439	-0.042	-0.008
30	0.011	1.277	0.582	-0.352	0.001	-0.605	-0.051	-0.020
40	0.011	1.749	0.800	-0.464	0.001	-0.762	-0.064	-0.024
50	0.011	2.248	1.003	-0.566	0.001	-0.923	-0.076	-0.028

Anomaly Detection to Identify Inaccurate Responses from Per-Student Gradients

Procedure

The same conditions and parameters were used, except for the number of schools and the proportion of contaminated rows. LOF and ABOD algorithms were fit on both gradients and the raw scores of a single 100 by 10 school response matrix. An additional simple extreme response indicator using the absolute deviation of each student's score is included as a baseline comparison. The proportion of contaminated rows was fixed at 0.3.

Performance was measured using AUC given the objective was to identify inaccurate responses, and identified rows could be compared against their true labels.

Results

Table 3: Detection AUC by contamination type (p = 0.3)

Type	LOF (grad)	ABOD (grad)	ABOD (raw)	Row-sum rule
Extreme (all-1s)	0.029	0.029	0.957	0.786
Straightlining	0.384	0.356	0.330	0.790
Random	0.693	0.770	0.747	0.339

LOF and ABOD are density/local methods, thus a large number of similar points induces the opposite effect on AUC scoring. A majority of similar points in the case of the extreme responses and straightlining would appear as "typical" to these algorithms, and any response that is the minority would be treated as an outlier whether the answer is scored correctly or not.

We can confirm this is the case when we compare the average LOF and ABOD scores for clean rows against the known contaminated rows. Contaminated rows scoring lower than clean rows is the density inversion that drives AUC below 0.5.

Table 4: Mean LOF score, clean vs contaminated rows.

Scenario	Clean rows	Contaminated rows
Extreme, gradient	Inf	1.000
Straightlining, gradient	1.345	1.185

Due to this effect, the simple absolute deviation scoring method is much more capable in addressing this specific pattern of inaccurate responding.

However, where the LOF and ABOD algorithms outperform the absolute deviation scoring are the random responding cases. ABOD seems to marginally outperform LOF on detecting random respondents when fitted on the gradients, as confirmed across 20 different seeds below.

Table 5: Detection AUC on random responding across 20 seeds (gradient features).

Statistic	LOF	ABOD	Row-sum rule
Mean AUC	0.654	0.742	0.361
SD	0.038	0.043	0.050

Discussion and Future Directions

The original study has already demonstrated the effects of extreme response rows on the bias and MSE of the item parameters- that these values will increase as more rows are contaminated, except for the bias of the difficulty parameter which grows increasingly negative. Straightlining follows a similar pattern at much smaller magnitude, but random responding departs from it: its discrimination bias carries the opposite sign, deflating rather than inflating the parameter.

The detectability of these inaccurate responses is dependent on the geometry of the contamination patterns, not so much the severity of the contamination. This causes a misalignment between the contamination worth removing and the contamination we can detect: truly harmful cases are trivially caught by an absolute deviation method (Row sum $AUC \approx 0.79$ for extreme responses and straightlining), while the density-based gradient methods only catch random responding, which is the least harmful type.

The gradient features did not clearly outperform detection on the raw responses, which is expected under binary scoring where the two are nearly redundant, and is a further reason the polytomous case is worth testing.

The next steps from here should be an examination of the net-effect of detecting and removing contaminated rows. Would introducing the probability of false-positives in order to remove inaccurate/contaminated responses be worth it over simply ignoring their effect on estimate accuracy? Another potential avenue could be to test the efficacy of these algorithms against other potential forms of inaccurate responding, such as cheating, sharing responses, and guessing concentrated on difficult items. Furthermore, these instances could be further examined in a partial credit scoring test, rather than dichotomous scoring.